

Mathematics Extension 2

**General
Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Centre Number and Student Number at the top of this page and on the Question 11 Writing Booklet attached

**Total marks:
100****Section I – 10 marks** (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7–14)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

1 What are the square roots of $3 - 4i$?

A. $1 - 2i$ and $-1 + 2i$

B. $1 + 2i$ and $-1 - 2i$

C. $2 - i$ and $-2 + i$

D. $-2 - i$ and $2 + i$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (10 marks)

- (a) The complex numbers w and z are given by $w = 2e^{i\pi/6}$ and $z = 3e^{i\pi/6}$. **2**

Find the modulus and argument of wz .

.....

.....

.....

.....

.....

- (b) The complex number z is given by $x + iy$.

Find, in Cartesian form:

(i) z^2 **1**

(ii) z^{-1} **2**

End of Question 11